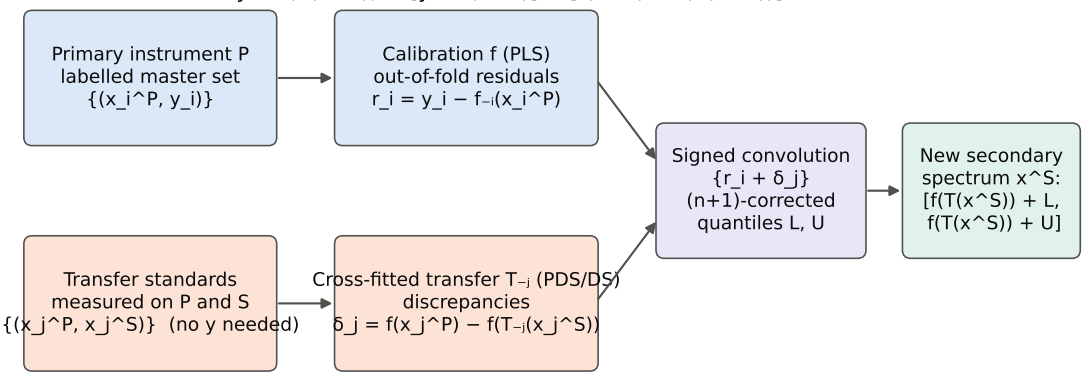
y − f(T(x^S)) = [y − f(x^P)] + [f(x^P) − f(T(x^S))] = r + δ
Primary instrument P
labelled master set
{(x_i^P, y_i)}
Calibration f (PLS)
out-of-fold residuals
r_i = y_i − f−i(x_i^P)
Transfer standards
measured on P and S
{(x_j^P, x_j^S)} (no y needed)
Cross-fitted transfer T−j (PDS/DS)
discrepancies
δ_j = f(x_j^P) − f(T−j(x_j^S))
Signed convolution
{r_i + δ_j}
(n+1)-corrected
quantiles L, U
New secondary
spectrum x^S:
[f(T(x^S)) + L,
f(T(x^S)) + U]